

Muhammad Abdullah

Date of birth: 13/11/2005 | **Nationality:** Pakistani | **Phone number:** (+92) 3094157102 (Mobile) | **Email address:** abdullahse2211@gmail.com | **LinkedIn:** Muhammadabdullahse2211 | **GitHub:** abdullahse2211 | **My_Portfolio :** Muhammad Abdullah |

Address: 53400, Lahore, Pakistan (Home)

EDUCATION AND TRAINING

02/11/2022 – 10/08/2026 Lahore, Pakistan

BACHELOR OF SOFTWARE ENGINEERING Lahore Garrison University, DHA, Lahore

Machine Learning, Probability & Statistics, Stochastic Processes, Linear Algebra, Calculus, Data Structures & Algorithms, Big Data Analytics, Operational Research.

Final grade 3.0/4 | **Level in EQF** EQF level 6 |

Thesis RetinaXplain: A Hybrid CNN–Vision Transformer Framework for Diabetic Retinopathy Classification

WORK EXPERIENCE

AI/ML ENGINEER (FULL-TIME) – AILYTICA – 01/09/2025 – Current – LAHORE, PAKISTAN

Medical Vision & Feature Segmentation: Designed and deployed hybrid CNN-Transformer models (Swin-UNet, MedSAM adaptations) for automated 2D/3D feature extraction and multi-class image segmentation, achieving sub-second GPU latency and high Dice similarity scores across target datasets.
MLOps, Quantization & Inference Infrastructure: Engineered end-to-end MLOps pipelines utilizing vLLM and TensorRT-LLM to serve quantized models (AWQ/QAT), reducing VRAM consumption by 45% while doubling inference throughput across Docker-containerized microservices.
Edge Vision & Real-Time Tracking: Built real-time visual tracking and spatial feature-extraction microservices using OpenCV, TensorRT, and PyTorch, optimizing processing speed to 30+ FPS for deployment on resource-constrained embedded nodes.
GraphRAG & Knowledge Retrieval: Developed enterprise GraphRAG architectures integrating Graph Neural Networks (GNNs) with vector databases (Qdrant), improving retrieval contextual accuracy by 35% on complex structured datasets.
Dataset Engineering & Model Benchmarking: Built automated data preprocessing, synthetic augmentation, and benchmarking scripts in PyTorch for deep learning backbones (UNet, ResNet, EfficientNet), establishing baseline metrics for downstream production tasks.

AI/ML ENGINEER (INTERN) – AILYTICA – 01/06/2025 – 30/08/2025 – LAHORE

Data Engineering & Preprocessing: Built automated data preprocessing and synthetic augmentation pipelines in PyTorch for deep learning models.
Baseline Architecture & Scripting: Developed initial benchmarking scripts for U-Net, ResNet, and EfficientNet architectures.
Evaluation Metrics & Workflows: Established baseline evaluation metrics for downstream computer vision tasks and assisted with dataset engineering workflows.

PUBLICATIONS

2026

RetinaXplain: Explainable Deep Learning Pipeline for Diabetic Retinopathy Severity Grading

Developed an automated deep learning pipeline for diabetic retinopathy grading using Swin-T achieving 85.27% accuracy.
Integrated multi-model inference and automated diagnostic report generation, documented in a 72-page technical report.

Muhammad Abdullah, Omer Aftab (2026). RetinaXplain: Explainable Deep Learning Pipeline for Diabetic Retinopathy Severity Grading. Manuscript in Preparation.

Authors: Muhammad Abdullah, Omer Aftab | **Journal Name**: Manuscript in Preparation | **Publisher**: Lahore Garrison University AI Lab

ACADEMIC & AI LAB RESEARCH PROJECTS

01/01/2026 – 22/05/2026

RetinaXplain - Hybrid CNN-Vision Transformer Framework for Diabetic Retinopathy Classification

Final Year Project-*Sole AI Researcher and Developer*
Data Pipeline & Preprocessing: Processed 3,662 APTOS images across 5 DR classes using circular cropping and Ben Graham normalization, expanding data to ~16,000 augmented instances.
Model Training & Algorithmic Features: Trained and evaluated Swin-T (85.27% accuracy, 0.7216 macro-F1), EfficientNet-B3, and ViT-B/16, engineering weighted soft-voting ensembles, clinical label remap,& agreement analysis.
App Deployment & Documentation: Built a multi-page web app featuring real-time DR diagnostic reports, probability visualizers, session analytics, and downloadable JSON records; authored a 72-page technical report.

Links https://github.com/abdullahse2211/diabetic-retinopathy-classification | https://youtu.be/MnS1vS5ZLqU?si=td3M1nW6vMmZRVf5

07/01/2026 – 08/03/2026

BioNAFNet - Low-Light Image Enhancement | Student Deep Learning Researcher

Novel Architecture & Loss Formulation: Designed a bio-inspired enhancement network trained from scratch with a composite Charbonnier, SSIM, color, and Weber-Fechner perceptual loss, surpassing 9 baselines with 28.67 dB PSNR on LOLv2 Real.
Knowledge Distillation & Acceleration: Compressed model parameters from 154M to 24M via knowledge distillation (6.4x compression, 3.9x speedup) with zero perceptual degradation, enabling the student model to surpass the teacher by +0.29 dB.

01/12/2025 – 02/02/2026

ThalCare - Alpha Thalassemia Classification | Student ML Researcher

Clinical Feature Engineering: Engineered 15 diagnostic features combining 9 CBC/HPLC parameters with 6 clinical indices (Mentzer, Shine & Lal, Green & King), elevating Silent Carrier diagnostic F1 to 0.92–0.955.
Robust Multi-Class ML Benchmarking: Evaluated 6 ML architectures (hybrid CNN-Transformer, 1D-CNN, SVM, RF, LR) for 5-class differential diagnosis, achieving 90.60% balanced accuracy and 0.906 weighted F1 under injected clinical noise and 10% feature dropout.

Links https://github.com/abdullahse2211/ThalCare.git | https://youtu.be/ciiGgdlZ4d4?si=_xKxSkJOZ9t3njEA

LEADERSHIP, HACKATHONS & MENTORSHIP

04/03/2026 – 05/03/2026

Flagship AI Hackathon Director

Spearheaded and personally directed the department's flagship AI Hackathon, orchestrating technical infrastructure, real-time submission evaluation pipelines, and jury coordination for 50+ participants across 15 teams.

04/01/2026 – 31/05/2026

Peer Mentoring & Technical Support

Actively mentored junior lab members on advanced PyTorch workflows, GPU optimization techniques, and Git-based collaborative development.

02/09/2025 – 08/06/2026

Departmental Event Organization

Assisted in organizing technical guest lectures featuring software industry executives and supported student exposure visits to local tech firms.

SKILLS

AI, Vision & Frameworks:

PyTorch, TensorFlow, OpenCV, scikit-learn, MedSAM, Swin-Transformer, UNet3+, Diffusion, RAG

Systems, MLOps & GenAI:

VLLM, TensorRT, Triton, Docker, 4-bit AWQ, Qdrant, GraphRAG, Qwen2-VL, LlamaIndex, LoRA, BoTorch

Languages & Tools:

Python, C++, SQL (MySQL), MATLAB, NumPy, Pandas, Matplotlib, Streamlit, Git, GitHub, Overleaf

CERTIFICATIONS

PERSON VUE CERTIPORT, 13/01/2025
Information Technology Specialist: JavaScript

International industry certification in JavaScript. Credential ID: mTp4-DwVM

Link https://verify.certiport.com/

National Vocational & Technical Training Commission (NAVTTC), 16/12/2024
JavaScript Full-Stack Development (MERN)

University of Engineering and Technology, Lahore National Centre of Artificial Intelligence, 08/12/2023
Prompt Engineering - Certificate of Completion

IBM; Coursera, 18/11/2023
IBM Web Development

Link https://coursera.org/verify/E8X42H6HASJL

CONFERENCES AND SEMINARS

11/11/2024 – 12/11/2024 Lahore
National Symposium on Artificial Intelligence & Medical Imaging

12/08/2025 – 14/08/2025 Lahore
Computer Vision & Edge Intelligence Summit

17/06/2025 – 18/06/2025
Advanced MLOps & Large Language Models Workshop

16/11/2025 – 16/11/2025 Lahore
Data Science & GraphRAG Technical Seminar

02/06/2026 – 02/06/2026 Lahore
Next-Gen AI Research & Vision Systems Colloquium

19/10/2024 – 19/10/2024 Lahore
National Tech Industry Leadership & AI Conference

15/10/2024 – 15/10/2024 Lahore
Information Technology Career Symposium 2024

24/09/2024 – 25/09/2024 Lahore
14th National Youth Peace Festival 2024

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s): **ENGLISH**